# Technical Design Report: Virtualisation Cluster Upgrade for electric-petrol.ie

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## Introduction

This report outlines the technical steps and procedures required to upgrade the existing two-host virtualisation setup at each branch site to a three-node Hyper-V failover cluster with shared storage. This upgrade will ensure high availability (HA) in case of a single server failure, significantly improving the resilience of critical services for electric-petrol.ie. The proposed solution will consist of the following key enhancements:

### Expanding to a Three-Node Cluster:

Adding a third host server allows the customer to create a more robust and redundant cluster.

### Implementation of Clustered Shared Storage:

Utilising either Storage Spaces Direct or an alternative shared storage solution to facilitate seamless failover.

### Enhanced High Availability (HA):

Guaranteeing continuous operation for critical Virtual Machines (VMs) even if one host should fail.

### Improved AAA Integration:

Re-engineering Authentication, Authorisation, and Accounting (AAA) services for tighter and more streamlined integration with the electric-petrol.ie domain.

### Additional Technical Benefits at No Extra Cost:

Leveraging existing Windows Server Datacentre licensing to provide features like live migration, improved redundancy, and streamlined management.

## Current Infrastructure Overview

### Existing Host Servers (Dell R440s)

The current virtualisation infrastructure at each branch consists of two Dell R440 servers with the following specifications:

- CPU: Dual Intel Xeon Silver (specific model if known)
- RAM: 256GB (configured with 16GB DIMMs)
- Storage:
    - OS Drive: 2 x 240GB SSD (RAID 1 configuration for operating system)
    - Data Drive: 4 x 1TB HDD (RAID 5 configuration for virtual machine data)
- Networking:
    - 2 x 1Gb/s onboard Ethernet 2 x 10Gb/s
    - Mezzanine Ethernet Adapters
- Hypervisor: Windows Server 2019 Datacentre
- Virtual Machines (VMs) allocation per host:
    - Domain Controller, 4GB RAM
    - File Server, 8GB RAM
    - SQL Server, 24GB RAM
    - Ubuntu Syslog, 8GB RAM (Allows monitoring, troubleshooting and maintaining system health)

## Customer Requirements

To address the growing need for system resilience and performance, electric-petrol has outlined the following key requirements:

1. *Add a Third Host Server for Redundancy:*
   Increase the number of active hosts to enhance fault tolerance.
2. *Implement Clustered Shared Storage (Hyper-V HA):*
   Establish shared storage accessible by all cluster nodes to enable high availability features.
3. *Ensure Operation Continues if One Server Fails:*
   Critical services must remain operational without interruption in the event of a single server failure.
4. *Re-engineer AAA into electric-petrol.ie Domain:*
   Modernise and integrate authentication, authorisation, and accounting services with the central domain.

## Proposed Upgrade Plan

This section details the comprehensive plan for upgrading the virtualisation cluster, encompassing new server specifications, storage solutions, network adjustments, Hyper-V cluster configuration, and AAA integration.

### A. New Server Specifications (Dell R450 Recommended)

To ensure seamless integration and compatibility, the new third server will largely match the existing infrastructure, with strategic minor upgrades for future growth and performance.

| Components | Specs | Reasoning |
|---|---|---|
| Device Model | Dell PowerEdge R450 | Latest equivalent of R440, ensuring compatibility and support. |
| RAM Allocated | 256GB (16GB DIMMs) | Matches existing nodes for consistent performance. |
| Operating System Storage | 2 x 480GB SSD (RAID 1) | Larger OS drive for enhanced capacity and future growth. |
| Data Storage | 4 x 2TB HDD (RAID 5) or 2 x 1TB SSD (RAID 1) | Increased capacity/performance for VM data. |
| Networking | 2 x 1Gb/s + 2 x 10Gb/s (mezzanine) | Matches existing setup, providing necessary bandwidth. |
| Hypervisor | Hypervisor Windows Server 2019 Datacentre | Maintains licensing consistency and familiar management. |

## B. Storage Upgrade (Clustered Shared Storage)

Implementing clustered shared storage is fundamental to enabling Hyper-V High Availability. We recommend the following options:

### Option 1:

Storage Spaces Direct (S2D) (if budget allows for SSDs) Requires at least two SSDs per node to serve as a caching tier. Provides resilient, distributed storage across the cluster nodes. Offers excellent performance and scalability.

### *Option 2:*

External SAN/iSCSI Storage (if budget is constrained) A Dell MD3420 could be added as a dedicated external shared storage solution. Connects to the cluster nodes via iSCSI. A more traditional approach to shared storage.

## C. Network Adjustments for High Availability

Optimizing the network configuration is crucial for the performance and reliability of the failover cluster:

- Dedicated 10Gb/s Network: A dedicated 10Gb/s network will be provisioned for cluster heartbeat and storage traffic, minimizing latency and maximizing throughput.
- VLAN Segmentation: Implementing VLAN segmentation for management, storage, and VM traffic will enhance security, isolate traffic, and improve overall network performance.

## D. Hyper-V Cluster Configuration

The Hyper-V cluster will be meticulously configured to maximise high availability and operational efficiency:

1. *Enable Failover Clustering:*

The Failover Clustering feature will be enabled on all three nodes.

2. Configure Cluster Shared Volumes (CSV):

   Cluster Shared Volumes will be set up to allow all nodes to concurrently access VM storage.

3. *Set VM Priority:*

   Virtual Machine priorities will be configured (e.g., DCx and DBx as high availability) to ensure critical VMs restart first in a failover scenario.

4. *Enable Live Migration:*

   Live Migration will be enabled to allow for seamless, non-disruptive transfers of VMs between cluster nodes for maintenance or load balancing.

### E. AAA Integration with electric-petrol.ie

The Authentication, Authorization, and Accounting (AAA) services will be re-engineered for robust integration with the electric-petrol.ie Active Directory domain:

- *Migrate AAA Services:*

  AAA services will be migrated to Active Directory-integrated RADIUS (Network Policy Server - NPS) for centralised management and enhanced security.

- *Ensure Time Synchronization:*

  Time synchronization across all servers will be maintained via the existing Galleon NTS-6002-GPS to prevent authentication issues.

## High Availability & Failover Strategy

The core of this upgrade is to establish a robust high availability and failover strategy:

- *Automatic VM Restart:*

In the event of a host failure, Virtual Machines will automatically restart on the surviving nodes, ensuring continuous service availability.

- *Node Majority Configuration:*

The three-node cluster will be configured with a Node Majority setup, meaning that at least two nodes must be online and communicating for the cluster to work correctly. This grants more resilience against individual node failures.

- *Priority Settings:*

Critical VMs, such as the Domain Controller and the database server, will be assigned high priority settings to ensure they restart first during any failover event.

## Additional Technical Benefits at No Extra Cost

Using the most out of the existing Windows Server 2019 Datacentre licensing, the upgrade will bring several additional technical benefits without incurring extra software costs:

- *Live Migration:*

Seamlessly move running VMs between cluster nodes without any downtime, enabling flexible maintenance and load balancing.

- *Storage QoS (Quality of Service):*

Prevent individual VMs from monopolizing storage resources, ensuring consistent performance for all virtual machines.

- *Better Monitoring:*

Operating the Windows Admin Centre for free, centralised, and intuitive cluster management, simplifying administration tasks.

- *Backup Improvements:*

Implement Azure Backup Server (included with Datacentre licensing) for improved backup capabilities and integration with cloud-based recovery options.

## Implementation Steps

The upgrade will be carried out in a structured manner, adhering to the following key implementation steps.

The first step involves acquiring and physically deploying the new Dell PowerEdge R450 server. After this, existing hosts (in this case the R440s) will be upgraded by adding SSD cache drives if Storage Spaces Direct is used and updating firmware and drivers on all existing hosts to ensure compatibility and stability.

Networking configuration will then begin with the setup of a dedicated 10Gb/S network for storage and cluster heartbeat traffic. Next, the failover clustering feature will be installed, and the chosen shared storage solution will be configured. Existing virtual machines will then be migrated to the newly configured Cluster Shared Volumes. To validate high availability functionality, various failover scenarios will be thoroughly tested. Finally, the migration and integration of AAA services with active directory will be completed.

## Risks & Mitigations

While every effort will be made to ensure a smooth upgrade, potential risks have been identified, and strategies have been developed to handle them. For instance, if a storage performance bottleneck is identified, the plan is to implement SSD caching, especially if the choice is to go with Storage Spaces Direct (known as S2D) or go with the other option of selecting a high-performance SAN (Storage Area Network) solution.

To prevent congestion in the network a dedicated 10GBps network will be utilised for both storage and cluster heartbeat traffic, along with the segmentation of the network VLANs. Additionally, to manage a potentially boot storm regarding virtual machines during a failover process, staggered startup priorities will be implemented. Which can prevent overwhelming the system resources during a mass restart of virtual machines and even the host system in most cases.

## Implementation Strategy Across Multiple Sites

To effectively roll out the virtualisation cluster upgrade, a phased approach will be employed, tailored to each site's specific requirements, including existing infrastructure and the availability of spare hardware. This strategy minimises disruption and ensures a smooth transition to the new, highly available clusters. As shown below a standard pattern begins to form:

1 – Reuse spare servers where possible.

2. Build new cluster in parallel.

3. Cutover or switch to new network with failback.

4. Recover any older PowerEdge R440s.

### 8.1. Sites 1 and 2: Full Replacement with Parallel Build

For the first two branch sites, which currently rely on the Dell R440s, a complete refresh of the server hardware is recommended. This approach provides the most consistent and performant environment. New Servers, specifically three Dell PowerEdge R450s as specified above, will be purchased for each of these sites. The new three-node Hyper-V cluster will be built in parallel to the existing two-host setup to limit the amount of downtime required without affecting live operations.

Once this new cluster is verified, a carefully planned cutover of VMs from the older hosts to the new cluster will be performed. As a contingency plan, a failback plan will be implemented should the move not go according to plan/schedule. After the successful cutover to the new infrastructure, the older existing Dell PowerEdge R440s will be recovered and repurposed elsewhere.

### 8.2. Site 3: Phased Integration with Existing Spares

At the third site, existing spare servers will be leveraged to facilitate the upgrade, offering a cost-effective solution while still achieving high availability. Four spares are available at this site. These will be configured to form the new three-node cluster, which allows for testing with an extra node or as a quick replacement if its needed. Similar to the first and second sites, the new cluster will be built beside (in parallel) to the current setup, which means no downtime is experienced during the deployment phase of the switch.

A planned cutover will move the virtual machines to the new cluster, with a complete failback strategy in place for security and in the event should works go awry. Once the works are completed and the new setup is fully migrated and stabilised, the two existing Dell PowerEdge R440s will be recovered for new purposes.

### 8.3. Site 4: Optimised Integration with Fewer Spares

The fourth site will also utilise spare servers, but with a more streamlined approach given the availability of fewer spares.

- *Reusing Spares:*

With three spare servers on hand, these will be directly configured to form the new three-node Hyper-V cluster.

- *Build in Parallel:*

The new cluster will be built in parallel to the existing environment, ensuring minimal disruption to ongoing operations.

- *Cutover with Failback:*

A controlled cutover process will migrate VMs to the new cluster, backed by a robust failback plan.

- *Recover R440s:*

Once the new cluster is fully operational and stable, the two existing Dell R440s will be recovered.

## Conclusion

This comprehensive virtualisation cluster upgrade for electric-petrol.ie represents a significant enhancement to the reliability, performance, and manageability of critical IT services. Upon completion, the new infrastructure will provide:

- *High Availability:*

The ability to survive a single node failure, ensuring uninterrupted service for critical applications.

- *Better Performance:*

Improved performance through optimised shared storage and dedicated 10Gb networking.

- *No Extra Licensing Costs:*

Leveraging existing Windows Server Datacentre benefits to deliver advanced features without additional software expenditure.

- *Simplified Management:*

Streamlined administration with the help of the Windows Admin Centre.

*Next steps:*

The recommendation would be approving the server purchase and scheduling a suitable downtime window for the migration process from the old infrastructure to the new. This downtime is vital to ensuring a seamless transition from the old infrastructure to the new one when moving virtual machines and services, while minimising disruptions to electric-petrols' operations.

During this downtime window, existing virtual machines will be moved to the newly configured cluster shared volumes and the rigorous testing of failover scenarios will be conducted to validate the clusters high availability functionality.